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HELIPOINT DETECTOR

Rooftop Helipad Detection with Computer Vision, Satellite Images, and YOLO

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Machine Learning and Computer Vision

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1. Executive Summary

The Helipoint Detector project consists of a computer vision solution developed to detect helipads on rooftops in the city of São Paulo using high-resolution satellite images. It is an end-to-end system, covering everything from data collection and preparation to model training, evaluation, and final analysis.

The value of the project lies not only in the trained model, but mainly in the construction of a structured pipeline capable of transforming raw geographic coordinates into an annotated and reusable dataset. In this way, the work demonstrates not only modeling capability, but also mastery of data engineering, image curation, annotation standardization, and performance evaluation.

The final model achieved very strong results, especially considering the small size of the training set. The metrics obtained show that the detector was able to reach a high level of precision, recall, and mAP, validating the quality of the dataset that was built and reinforcing the idea that, in artificial intelligence projects, data quality is decisive for the success of the system.

2. Introduction

Automatic object detection in aerial and satellite images is an important application of computer vision, especially in complex urban scenarios. In the case of this project, the focus is on identifying helipads installed on top of buildings, structures that have operational and urban relevance, but whose manual detection at scale would be slow and difficult to maintain.

Helipoint Detector was conceived as an applied academic project, with the purpose of demonstrating the complete development cycle of an artificial intelligence solution aimed at geospatial imagery. Thus, the project goes beyond simply training a model and covers all the steps required to transform a real problem into a functional technical pipeline.

In addition, the task presents considerable visual difficulty. Helipads may vary in shape, color, paint wear, background contrast, shadow incidence, and degree of occlusion. In many cases, patterns present in swimming pools, sports courts, terraces, or other urban structures may be confused with the characteristic marking of a helipad, which makes the problem especially interesting from the perspective of classification and detection.

3. Problem Definition

The central problem of this work is the automatic detection of helipads on rooftops by means of satellite images obtained from public sources and processed with computer vision techniques. In a metropolis such as São Paulo, this task is challenging due to the high urban density, architectural diversity, and the presence of multiple similar visual patterns.

From a technical point of view, the challenge is not limited to visually recognizing the letter “H” or the typical geometry of helipads. It is also necessary to deal with variations in scale, orientation, lighting, shadow, low local resolution, visual noise, and overlap among urban elements. These factors increase the likelihood of false positives and false negatives, requiring a well-built dataset and a robust model.

Solving this problem means transforming unstructured images into structured geospatial information. This can support future applications in urban mapping, infrastructure monitoring, aerial navigation support, logistics studies, and automated visual inspections.

4. Objectives

The general objective of the project is to develop a complete system for detecting helipads in satellite images, using an approach based on convolutional neural networks and models from the YOLO family.

Among the specific objectives, the following stand out:

- Build an original helipad dataset for São Paulo.
- Collect and organize geographic coordinates of the targets.
- Obtain high-resolution satellite images.
- Perform manual annotation of the objects.
- Train an object detection model.
- Evaluate the model through quantitative metrics and qualitative analysis.
- Verify the system’s ability to generalize to areas not seen during training.

The project also aims to demonstrate that careful data creation is one of the main success factors in applied artificial intelligence. In this sense, the work functions simultaneously as a technical experiment, an academic case study, and a practical AI engineering portfolio.

5. Context and Background

This project is situated in the context of computer vision applied to urban geospatial imagery. The use of orbital or aerial data to detect specific structures is a relevant line of research and practical application, especially in large cities, where manual visual monitoring is costly and not very scalable.

In the academic environment in which the work was developed, there was a requirement to build an original dataset, document the pipeline, and demonstrate model training and validation. This guided the project toward an approach strongly oriented toward reproducibility, artifact organization, and clear separation between data discovery, preparation, annotation, training, and analysis.

From a technological point of view, the project relies on tools such as FlightMarket, Selenium, ESRI World Imagery, Roboflow, Jupyter notebooks, Google Colab, Ultralytics YOLO, and PyTorch. This combination made it possible to structure an accessible technical flow that was nevertheless robust enough to generate consistent results.

6. Solution Architecture

The solution architecture was organized into modules in order to allow methodological clarity and process repeatability. The flow starts from the discovery of helipads and their coordinates, moves through image acquisition, dataset preparation, object annotation, model training, and reaches the evaluation and inference stage.

This modular organization is important because it separates responsibilities and facilitates project evolution. Instead of concentrating all the logic in a single script, the system was designed as a sequence of complementary components, each responsible for a specific stage in the solution life cycle.

From a conceptual point of view, the architecture can be summarized into seven layers: geospatial discovery, image collection, data processing, annotation, training, evaluation, and results presentation. This structure provides a good foundation for future extensions, including web applications and flows closer to MLOps.

7. Data Pipeline

The data pipeline begins with the collection of helipad coordinates, obtained from external registration and query sources. These coordinates are organized into structured files and then converted into geographic regions of interest.

From these regions, satellite images are downloaded in tile format, later organized by neighborhood, cropped, assembled, and filtered. This stage is fundamental because it defines the visual quality of the training dataset and directly influences the detector's final performance.

After image acquisition, manual curation of the material takes place, removing inadequate examples and selecting the most useful crops. Next, the images are annotated with bounding boxes, resized to the training format, subjected to data augmentation, and divided into training, validation, and test sets.

This pipeline shows that the project does not depend only on the model, but on a well-defined processing chain. The quality of the detector is a direct consequence of the quality of the data flow.

8. Backend Architecture

Although the project does not have a traditional backend based on microservices, it presents a functional architecture centered on scripts, notebooks, and intermediate files. This approach is suitable for the academic scope, as it favors simplicity, traceability, and ease of experimentation.

The main components of this layer include scripts for coordinate collection, geographic transformation, image acquisition, exploratory analysis, model training, and inference execution. Each part of the flow generates specific artifacts, which are stored in organized directories for later inspection.

It is, therefore, a backend oriented toward batch processing and artifact generation rather than permanent online services. Even so, this structure is sufficient to demonstrate the complete pipeline with technical coherence.

9. APIs and Integrations

The project uses different external sources and platforms to compose its processing chain. Among them, satellite image services, annotation platforms, and training frameworks stand out.

Integration occurs mainly through file exchange, automated queries, dataset exports and imports, rather than through a complex internal API architecture. This design is coherent with the objective of the work, which is to demonstrate the technical cycle of the solution without unnecessarily increasing deployment complexity.

Even so, the experience shows how data from different origins can be combined into a unified flow. This integration between external sources and specialized tools is one of the project's most important aspects.

10. Artificial Intelligence Components

The artificial intelligence core of the project is a single-class object detector, trained to locate helipads in satellite images. The chosen approach is based on the YOLO family, recognized for its good balance between speed and performance in detection tasks.

The use of YOLO is appropriate because the problem requires not only classifying the presence of the object, but also locating its exact position in the image by means of bounding boxes. For this, training was carried out with data prepared in the YOLO standard, including images and corresponding label files.

In addition to the choice of architecture, the project also includes evaluation of classic detection metrics such as precision, recall, mAP@50, and mAP@50–95. This allows for a more rigorous analysis of the model's behavior and its spatial quality.

11. Dashboard and Analytics Layer

The project's analytical layer is based mainly on notebooks and artifacts generated during training. There is no formal corporate dashboard, but there is a sufficient structure to inspect metrics, charts, and visual outputs of the model.

The analyses include loss curves, the evolution of precision and recall, mAP curves, and qualitative inspection of predictions. This material makes it possible to understand both the learning dynamics of the model and its successes and limitations in concrete cases.

Although this approach is more handcrafted than a traditional executive dashboard, it serves well the role of supporting technical interpretation of the results. In academic work, this type of analytical layer is usually sufficient to demonstrate methodological maturity.

12. Chatbot and Conversational AI

The project does not implement a chatbot or a conversational AI system. Its focus is entirely on computer vision and object detection in geospatial images.

Even so, in a future evolution, it would be possible to add a conversational interface to support users in interpreting the results. A conversational agent could, for example, explain the confidence level of a detection, summarize metrics, or assist in the human review process of the images.

In its current state, however, this functionality is not part of the scope of Helipoint Detector and does not affect the results presented.

13. Security

As this is an academic and experimental project, the solution does not present a complete corporate security architecture. However, there are implicit concerns related to the use of external platforms, local credentials, notebook environments, and artifact sharing.

In projects of this nature, good practices include dependency control, care with exposure of access keys, repository organization, and review of published content. Although they are not the main focus of the work, these precautions are important to ensure traceability and reduce operational risks.

In a possible evolution toward a production environment, it would be advisable to incorporate secret scanning, permission control, dependency validation, and more formal mechanisms for artifact integrity.

14. Governance and Responsible AI

The project demonstrates important governance concerns by establishing annotation criteria, data curation, separation between training and validation, and testing in unseen regions. These methodological decisions help reduce hidden biases and make the results more reliable.

The presence of qualitative analysis is also relevant from the perspective of responsible AI, because it prevents evaluation from being restricted only to numerical metrics. Observing where the model succeeds, where it fails, and why it fails is fundamental to understanding its real limits.

As a future improvement, it would be interesting to further formalize dataset lineage, experiment tracking, and documentation of model versions. This would increase the capacity for auditing and comparison across runs.

15. Scalability and Performance

The current solution presents good scalability within the academic context, mainly because it automates important stages of geospatial discovery and data preparation. This reduces manual work and facilitates expanding the project to new neighborhoods or sets of images.

However, for a broader production scenario, additional mechanisms for orchestration, monitoring, model versioning, drift control, and experiment management would be necessary. In other words, the project already has a solid foundation, but it would still need maturation to operate as a large-scale product.

With regard to performance, the main potential bottlenecks are in data acquisition and preparation, human annotation time, and the computational cost of training. The use of GPU in Colab helped make the process more feasible.

16. Key Metrics

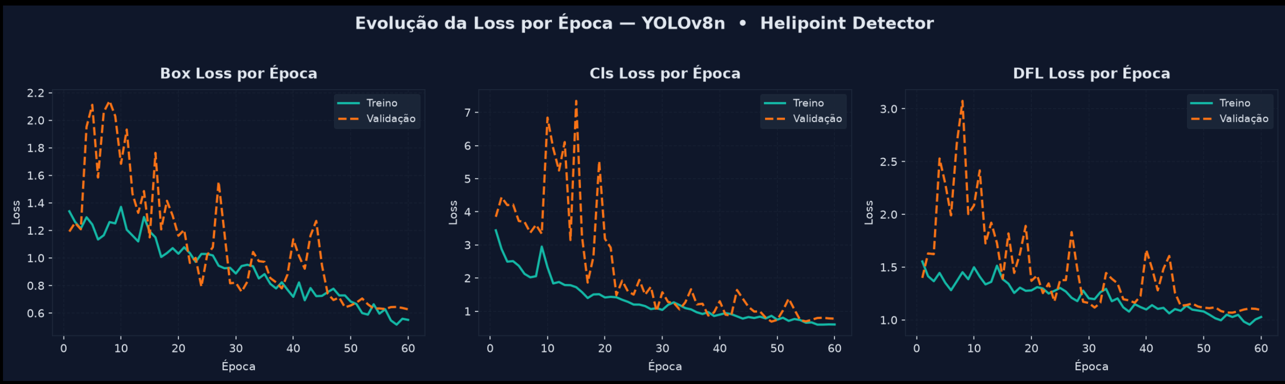
The results obtained by the model were quite expressive, especially considering the small size of the training dataset. The best overall performance occurred at epoch 54, while the final analyzed epoch was 60.

These numbers show that the model achieved an extremely high level of precision and excellent overall detection performance. The fact that mAP@50 remained at 0.994 both at the best point and at the final epoch indicates strong stability of the detector under the standard criterion. mAP@50–95, in turn, reinforces that the bounding boxes were also well adjusted under a stricter criterion.

Metric	Best Epoch (54)	Final Epoch (60)
Precision	1.000	0.992
Recall	0.963	0.971
mAP@50	0.994	0.994
mAP@50–95	0.881	0.841

17. Generated Charts

Figure 1 — Evolution of Losses by Epoch

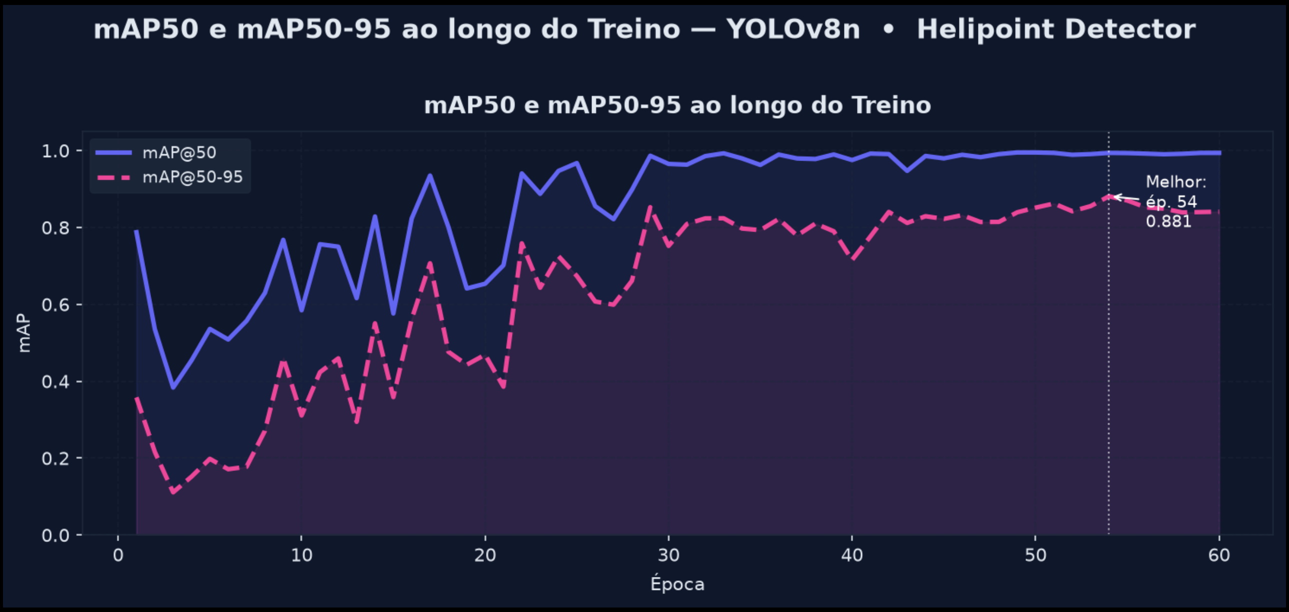


The loss curves indicate consistent learning throughout the 60 epochs. Both training and validation losses decrease progressively, suggesting that the model was able to learn useful patterns without evident signs of divergence between the two sets.

Figure 2 — Precision and Recall Throughout Training

The precision and recall curves show rapid performance evolution and stabilization at very high levels in the final stages of training. This suggests that the model was able to strike a good balance between detecting true objects and reducing false positives.

Figure 3 — mAP@50 and mAP@50–95



The mAP curves show excellent performance both under the more permissive criterion and under the stricter criterion. The peak at epoch 54 reinforces the perception that this was the optimal training moment for the available dataset.

18. Qualitative Analysis

Qualitative analysis is important because it complements numerical metrics with direct visual observation of the model's behavior. In object detection projects, this stage is essential to understand whether the system makes
mistakes because of geometric confusion, low visibility, visual noise, or contextual ambiguity.

Type	What to show	What to explain
Clear hit	Helipad detected with a well-adjusted box and high confidence	Show that the model recognizes the classic object pattern with precision
Challenging hit	Helipad partially covered or at an angle	Demonstrate the detector's robustness in less favorable situations
False positive	Structure similar to a helipad	Explain that certain urban patterns still generate visual confusion
False negative	Helipad not detected	Highlight the model's limits in cases of shadow, wear, or low contrast

19. Limitations

The main limitation of the project lies in the size and geographic scope of the dataset. Although the dataset was built in an original and careful way, it still represents a limited fraction of the possible visual diversity of urban helipads.

Another important limitation concerns architectural maturity. The project was well structured for academic and technical demonstration purposes, but it still does not have all the mechanisms that would be expected in a complete production solution, such as continuous monitoring, formal model versioning, and advanced operational governance.

There is also a limitation inherent to human annotation and visual ambiguity. Certain urban structures may be sufficiently similar to helipads to continue generating errors, even in well-trained models.

20. Future Improvements

Among future improvements, the first priority is to expand and diversify the dataset. Including more neighborhoods, different lighting conditions, new scales, and more varied negative examples tends to increase the detector's robustness.

Another important improvement is to formalize experiment tracking, allowing model versions, training parameters, and metrics to be compared more systematically. This would facilitate both academic analysis and the system's technical evolution.

It would also be interesting to improve the demonstration interface by incorporating batch upload, confidence filters, assisted review, and exportable reports. At a later stage, the project could evolve into a solution with explicit MLOps practices.

21. Results and Main Outcomes

The main outcome of the project was the construction of a functional end-to-end solution for detecting helipads in satellite images. This includes data collection, dataset creation, annotation, training, evaluation, and analysis of results.

In addition, the project demonstrated that it is possible to achieve very high performance even with a relatively small dataset, provided that data curation and annotation are carried out with quality. This is an important result from a methodological point of view.

Another relevant outcome was the production of a reusable pipeline. The created structure can serve as a basis for new experiments, geographic expansion, and adaptation to other types of urban objects.

22. Data Analysis and Insights

One of the project's main insights is that data quality has a decisive influence on model performance. In other words, the success obtained does not arise only from the choice of YOLO, but from the careful construction of the training set.

Another important insight is that generalization should be treated as a central evaluation criterion. Testing the model on unseen areas is essential to verify whether it has truly learned object patterns rather than merely specific characteristics of the training set.

It is also observed that model errors tend to be concentrated in visually ambiguous situations. This indicates that future improvements should prioritize enriching the set of hard negatives and challenging examples.

23. Decision Support and Recommendations

The main recommendation arising from this work is to prioritize the evolution of the data pipeline. Improvements in curation, visual diversity, quality control, and annotation criteria will likely have an impact as large as, or greater than, changes in the model architecture.

Another recommendation is to maintain the use of quantitative metrics combined with qualitative analysis. In computer vision projects, the exclusive reading of numerical indicators can hide relevant error patterns that only appear through visual inspection.

Finally, it is recommended to preserve evaluation in unseen neighborhoods as a permanent practice. This type of test is one of the best indicators of the system's robustness under conditions close to real use.

24. Conclusion

Helipoint Detector successfully fulfilled its objective of demonstrating the construction of a complete artificial intelligence solution aimed at detecting helipads on rooftops. The work showed mastery of data collection, geospatial engineering, image curation, annotation, training, evaluation, and interpretation of results.

The quantitative results confirm that the model achieved very high performance, even with a compact dataset. This reinforces the importance of dataset quality and validates the strategy adopted throughout the project.

Overall, the project has academic, technical, and practical value. It functions as concrete proof that the combination of well-prepared data, clear methodology, and rigorous evaluation can generate robust solutions even in challenging contexts.

25. Next Steps

The most natural next steps include expanding the dataset, consolidating experiment tracking, and strengthening the continuous evaluation structure. These measures should increase model robustness and facilitate comparisons between future versions.

It is also recommended to evolve the results presentation layer, either through a more complete web interface or through automatic reports with standardized visualizations and metrics. This would increase the project’s value for demonstrations, presentations, and future applications.

In the medium term, the solution may evolve into a broader geospatial system capable of identifying other types of urban structures, while maintaining the same methodological foundation employed in this work.